CSRV

CHARAN SAI REDDY VENKATAPURAM

Mechanical Design Engineer

Design & Build · Prototype to Production · Hardware Deployed in the Field



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Graduate Route visa to Feb 2027
Eligible for Skilled Worker sponsorship at the New Entrant rate

Professional Profile

Mechanical design engineer with six years' experience taking hardware from concept to product in the field — displays, cameras, readers and computing packaged into sheet-metal and plastic assemblies, prototyped by hand and maintained in service.

I come from a hands-on workshop background: I design for the manufacturing process actually available, and I build the jigs and fixtures needed to verify parts with the people who make them. I am comfortable owning the trade-offs between performance, cost, manufacturability and serviceability.

6 years	37	MSc
Design engineering	Units deployed in the field	Loughborough University

SolidWorks (CSWA · CSWP in progress) · Siemens NX 12 · Fusion 360 · ANSYS static structural · Sheet metal · Tube forming · Machining · FDM (PETG · PLA · ABS)

Currently building formal competence in GD&T (ISO GPS) and DFMEA (AIAG-VDA).

SELECTED WORK

Five projects — hardware built and deployed in the field, modular platform design, printed prototypes and welded vehicle structures

01

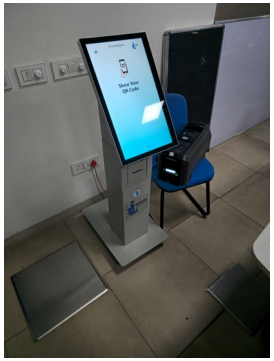
Self-Service Kiosk Programme



Government tender · 37 units deployed

02

Gen 4 Modular Kiosk Platform



Deployed at IISER Tirupati

03

Face-ID Terminal Enclosure



Three versions · 8 months

04

RFID Reader Enclosure



Concept development · ongoing

05

Vehicle Structures & Fabrication Fixtures

E-cargo two-wheeler prototype · ATV roll cage · 1:1 tracing jigs for external fabricators

Four generations of development

No prior products, documentation or domain precedent in-house. I benchmarked NCR, Diebold and Indian OEMs, set a form-factor and serviceability strategy, and drove the line from a retrofitted proof of concept to a deployed government product and then to a modular platform.



GEN 1 2018

The freelance proof of concept, retrofitted with A4 and card printing and a tablet PC — a photocopier-like form factor built to prove the function, and later shown publicly.



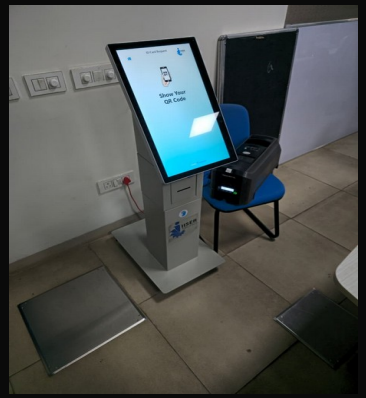
GEN 2 2019

A second proof of concept and the first fully bespoke enclosure — compact silhouette, 15-inch display, card and document handling — exhibited publicly.



GEN 3 2020

The first to ship — winner of the Election Commission of India tender, with a full-size Datacard printer in a quarter of the competing bid's footprint.



GEN 4 2025

A modular, stackable platform built from one repeated enclosure module and configured per deployment — now in service at IISER Tirupati.

01

TENDER AWARD & DELIVERY

Gen 3 — Election Commission of India, Tamil Nadu

- Packaged a full-size Datacard card printer into roughly a quarter of the competing entry's footprint — compactness and maintainability decided the award.
- The design went from concept to production in two months with an outsourced fabricator in Tirupati.
- Sourced and qualified the touchscreen supply chain independently, validating the vendor against their deployed public-transport installations before specification.
- Set up a temporary in-house build operation, defined the build sequence and trained two trade-qualified fitters — 37 kiosks entered service across 37 districts within a month.





Production units on the floor, and a deployed kiosk in service — camera head, touchscreen, RFID reader, card dispenser and document slot, with the backlit acrylic panel carrying voter information on the side face

37 units	1 month	1.6 mm
Deployed across 37 districts	Full field rollout	HR steel, powder-coated

Designing for the engineer who maintains it



Gen 3 with the service hood raised — the screen travels with the hood, opening access to the printer bay below and the cabling behind the display

- A front-swing door needs clear floor space in front of the unit — unavailable in a public queue. Instead the hood hinges at the top and opens flat, carrying the clamped touchscreen with it.
- One opening gives two service zones: the card printer below for ribbon and card reloading, and the display's rear ports, NIC and cabling presented as a level working plane.
- Welded U-clamp cable guides with service-length terminations keep looms controlled across a joint opened repeatedly over the unit's life.
- Display aperture cut to seat the touchscreen's own supplied gasket as the primary seal, back-filled with silicone as a secondary ingress barrier.
- Unused side-panel area turned into a backlit acrylic panel carrying voter information and usage instructions on the unit itself.

Accessibility drove the Gen 3 geometry

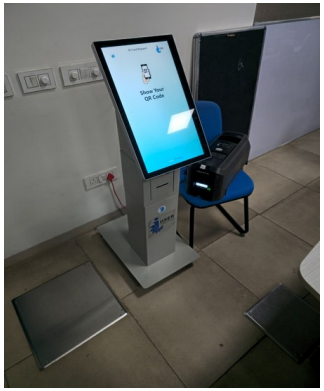
Highest touch point set at 48 in within the ADA forward-reach limit; screen rake fixed at 55° for a 4 ft to 6 ft 6 in user range. Two physical models built and compared before down-selecting.

02

GEN 4 MODULAR KIOSK PLATFORM

One enclosure module, many configurations

- Deployed at IISER Tirupati for student ID card issuance — QR-code request flow, card dispenser and printer integrated into the column.
- A repeated enclosure module stacks on a shared base with a separate screen-head assembly, so height and peripherals vary per deployment without redesigning the enclosure.
- Each module has its own lockable door and internal mounting rails — peripherals are serviced module-by-module, not by opening the full column.
- Separates what holds the component from what defines the product: the interface stays fixed as the contents change.







In service at IISER Tirupati and in an alternate screen-head configuration, with the module column opened — independent hinged doors, cam locks and internal mounting rails

03

FACE-ID TERMINAL ENCLOSURE

Three versions in eight months — each with a defined purpose, and each deployed to customer sites in small numbers



V1

Prove the function

The first version was built deliberately plain, to isolate assembly sequence, serviceability and user interaction before any styling. Its two-part architecture fixes and cables the ribbed wall plate first; the terminal then hangs on and key-locks.



V2

Restructure the architecture

Device retention moved onto an internal carrier frame, decoupled from the outer shell; the mains adapter is captured inside the wall plate, and a full-height light pipe carries status.

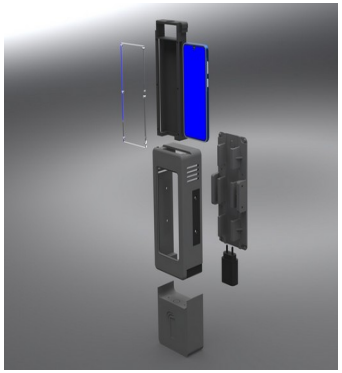


V3

Resolve the form

The product resolved into a single continuous slab, with the display flush in a close-tolerance aperture and every fastener, seam and service interface moved to the rear.

Architecture, materials and finish



V2 exploded view — shell, carrier frame, window, light pipe, and wall plate with captured PSU



V3 section view — integrated light pipe and internal wall structure

- Pull-out device tray with integrated cable guide and positive locking — the device is fitted and serviced from the top without disturbing the enclosure or its wall fixing.
- Charger seated and secured inside the wall plate, so the installed unit presents no accessible cabling or fasteners.
- 2 mm acrylic bonded into a seating recess as an impact and tamper barrier for deployment in educational institutions.
- PETG specified across the product family for RF transparency, alongside service temperature and print reliability.
- Print orientation and geometry controlled to achieve a fine textured finish direct from FDM with no post-processing.
- The future version is defined as a hybrid build — printed front touch-point with a bent and powder-coated aluminium rear structure; MJF-processed PA 12 under evaluation for outdoor and flame performance at volume.

04

RFID READER ENCLOSURE

Concept development — ongoing

- Five concept directions generated and down-selected through physical 3D-printed prototypes rather than screen review alone — assessing proportion, hand feel and wall-mounted presence.
- PETG selected over PLA for service temperature and over ABS for print reliability, availability and reduced fume hazard.
- RF transparency treated as a material requirement from the outset rather than a late correction.
- Chamfered perimeter, recessed fascia and debossed iconography developed in CAD alongside printed parts to judge surface treatment before committing.







Wall-mounted concept, CAD render and printed PETG prototype

E-cargo two-wheeler: concept to road-tested prototype

- Bespoke cargo frame with compound tube bends across multiple planes, packaged around a repurposed donor chassis and swing arm.
- Rear hub motor, front and rear disc brakes and an in-house thumb throttle integrated; brakes bled and the vehicle commissioned and road-tested.
- IS 2062 circular tube specified as an available proxy for AISI 1020 to validate geometry, fit-up and assembly; 4130 evaluated and rejected on cost.
- First-pass static structural FEA in ANSYS to confirm load paths, with mid-span forming variance accepted where end coordinates and height met specification.





SolidWorks assembly (left) and the prototype as built (right) — repurposed donor chassis, rear hub motor, disc brakes front and rear

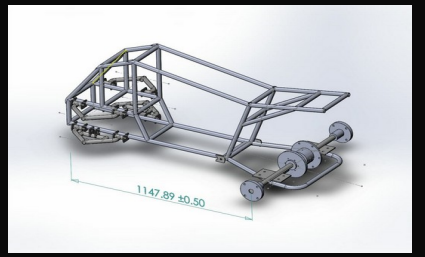
ATV roll cage and the 1:1 fixture method

Quad-TORC national design competition — Team Design Lead

- Roll cage design lead for a student off-road quad bike, competing at national level.
- AISI 4130 chromoly specified for variable, multi-axis off-road loading; static structural FEA plus front and side impact load cases.
- Wheelbase and frame envelope dimensioned to ±0.5 mm in CAD and verified against the welded structure during build.
- Disc brake flanges, wheel hubs and rear swing arm machined in-house; Maruti Alto master cylinder, knuckles, hub carriers and steering rack integrated into the bespoke chassis.







Frame CAD dimensioned to ±0.5 mm; the welded structure verified against it, and the completed vehicle

METHOD — 1:1 TRACING JIGS FOR EXTERNAL FABRICATORS

1 Plot 1:1 on A1 › **2** Timber former › **3** Mark bend start/end › **4** Sequence & axis › **5** Verify formed tube

Developed for geometry beyond the available cold die-bending capability — no tooling spend, reused and extended across two programmes four years apart.

CONTACT

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